



FABRIC STRUCTURE & DESIGN

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Commercial Fabrics

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FABRIC BASED ON **PLAIN** WEAVE

Georgette: Georgette is made with filament yarns. In georgette, the direction of the crepe twist (S or Z) for warp and weft yarns alternates. For example, even-numbered warp and weft yarns may be S-twist and odd-numbered yarns may be Z-twist. It can be a solid color or printed. It is very lightweight, drapes well, and is used in apparel. It was originally made of silk but now is often made from manufactured filament yarns.

Chiffon: It is made from fine, highly twisted filament yarns. Because of the tightly twisted crepe yarns, chiffon has excellent drape, very light weight, and although it is delicate in appearance, it is relatively durable. It can be a solid color or printed. It was originally made of silk but now often is made from manufactured filament yarns. Sheer evening dresses, blouses, lingerie, and other dressy apparel are constructed from this fabric.

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Voile: Voile is a sheer fabric made with high twist or voile twist spun yarns that are combed or worsted. It is a soft fabric with somewhat lower fabric count and has a distinctive two-ply warp and good drapability. It can be solid color or printed. Voile was originally a cotton or wool fabric, but it is now available with many fiber contents.

Organdy: It is the sheerest cotton fabric that is given a temporarily or permanently stiffened finish. Combed yarns contribute to its sheer appearance. Its sheerness and crispness are the result of an acid finish on lawn grey goods. Because of its stiffness and fiber content, it is very prone to wrinkling. It is used for curtains and summer-weight apparel. It is available in solid colors or prints. Fabric construction is similar to lawn fabric.

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Organza: It is the filament yarn counterpart to organdy, i.e. it is also a stiff sheer fabric made of filament yarns. It has a lot of body and a crisp hand. It is also used for curtains and for summer-weight apparel. It is available in solid colors or prints.

Lawn: It is a fine, opaque, lightweight, plain weave fabric usually made of combed cotton or cotton-blend (cotton/polyester). The fabric may be bleached, dyed, or printed. Lawn is similar to organdy fabric, but it does not receive the acid finish like organdy and, thus, remains opaque. Fabric construction: 70' x 100' / 80 x 80.

Batiste: It is an opaque, lightweight, spun yarn, plain weave fabric with a smooth surface. It is the softest of the lightweight opaque fabrics. When made of cotton or cotton/polyester, the yarns are usually combed. It can be made of all wool, silk, or rayon. Batiste is also similar to organdy fabric, but it does not receive the acid finish like organdy and, thus, remains opaque. Fabric construction is similar to lawn.

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Gingham: It is a yarn-dyed, plain weave fabric that is available in a variety of weights and qualities. It may be: Balanced or unbalanced. Made of combed or carded yarns. If two colors of yarns are used, the fabric is called a check or checked gingham. If three or more colors are used, the fabric is called a plaid gingham. It is usually made of cotton or cotton blends. Better quality fabrics are made with combed yarns. When another fiber is used, the fiber content is included in the name, for example, silk gingham. Fabric construction: 28' x 42' / 64 x 60 to 64 x 76 (carded) 28' x 42' / 84 x 76 to 88 x 84.

Chambray: It is a plain weave fabric, usually made of cotton, rayon, or blended with polyester. Usually, chambray has white yarns in the weft direction and colored yarns in the warp direction. Iridescent chambray is made with one color in the warp and a second color in the weft. It can also be made with stripes.

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Taffeta: It is a general term that refers to any plain weave filament yarn fabric with a fine, smooth, crisp hand. Unbalanced taffeta has a fine rib made by heavier filling yarns and more warp yarns. Faille taffeta has a crosswise rib made by using many more warp yarns than weft yarns. Moiré taffetas have an embossed water mark design. Balanced taffetas have warp and weft yarns of the same size. Iridescent taffeta has warp and weft yarns of different colors.

Madras: It is usually all cotton, and has a lower count than gingham. Madras gingham or madras shirting is a light to medium-weight, dobby weave fabric in which the pattern is usually confined to vertical stripes.

Cheese cloth: It is a light, sheer, plain-woven fabric with a very soft texture and a very low count. It may be natural colored, bleached, or dyed. If dyed, it may be called bunting and could be used for flags or banners.

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Crinoline: It is a stiff, spun yarn, plain weave fabric similar to cheese cloth, used in book bindings, hats and stiffening for apparel. It is heavily sized to serve as stiffening fabrics.

Buckram: It is heavily sized to serve as stiffening fabrics. Buckram is a heavy, very stiff, spun yarn fabric converted from cheese cloth gray goods with adhesives and fillers. It is used as an interlining to stiffen pinch pleated window treatment fabrics.

Gauze: It is a sheer, lightweight, low count, plain or leno weave balanced fabric made of spun yarns. It is often cotton, rayon, or a blend of these fibers. Gauze, with a higher count than cheese cloth, is used in theatrical costumes and medical dressings, as well as for blouses and dresses. Indian gauze has a crinkled look and is available in a variety of fabric weights.

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Ninon: It is a sheer, slightly crisp, lightweight, plain weave fabric made of filament yarns. The warp yarns are grouped in pairs, but it is not a basket weave fabric. It is widely used in sheer curtains and draperies. It is usually 100 percent polyester because of that fiber's resistance to sunlight, excellent resiliency, and easy washability.

Calico: It is a closely woven and print cloth of cotton or cotton blend with a small busy pattern.

Cambric: It is a fine, firm, starched plain weave balanced fabric with a slight luster on one side. It is difficult to distinguish from percale.

Percale: Percale is a smooth, slightly crisp, printed or plain colored fabric. It is a closely woven, plain weave of cotton or blended fibers, is made from yarns of moderate twist. Percale yard goods are generally carded, but percale sheets are finer and more luxurious in feel and are made of combed yarns. In percale bedsheets, counts of 160, 180, 200, and 250 yarns

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(warp plus weft) per inch are available. Percale is called calico if it has a small, quaint, printed design; chintz if it has a printed design; and cretonne if it has a large-scale floral design. When a fabric is given a highly glazed calendar finish, it is called polished cotton. When chintz is glazed, it is called glazed chintz. Glazed chintz is made in solid colors as well as prints. These fabrics are often made with blends of cotton and polyester or rayon. They are used for shirts, dresses, blouses, pajamas, matching curtains and bedspreads, upholstery, slipcovers, draperies, and wall coverings.

Muslin: It is a firm, medium to heavy weight, plain weave cotton fabric made in a variety of qualities. It generally woven from cotton or cotton blends, is made in both heavily sized, bleached qualities and in better grades for sheets and pillow cases. Any plain woven, balanced fabric of carded yarns ranging in weight from lawn to heavy bed sheeting may be called muslin. It is usually available in counts of 112, 128 or 140. Muslin is also a name for a medium weight fabric that is unbleached or white.

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Flannel: It is a light to heavy weight, plain or twill weave fabric. Flannel is a suiting fabric of woolen yarns that is napped. It is used for women's suits, slacks, skirts, and jackets. Flannelette is a light to medium weight, plain weave cotton or cotton blend fabric lightly napped on one side. It can be found as both balanced and unbalanced fabrics. It is available in several weights ranging from 4.0 to 5.7 oz/yd². It is described as flannel and is used for sheets, blankets, and sleepwear. Outing flannel is heavier and stiffer than flannelette, it may be napped on one or both sides. It is used for shirts, dresses, lightweight jackets and jacket linings. Some outing flannels are made with a twill weave. Both fabrics may be solid color, yarn dyed or printed.

Poplin: Poplin is a bottom-weight rib weave (heavy weight ribbed fabrics) is usually made from cotton or cotton blends, polyester/cotton blends are widely used. It is a medium to heavy weight, unbalanced, plain weave, spun yarn fabric that is usually piece-dyed. The weft yarns are coarser than the warp yarns. Poplin is similar to broad cloth, but the ribs are heavier and more pronounced because of larger weft yarns.

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Canvas: It is a heavy, firm, strong fabric made of cotton or acrylic and used for awnings, slip covers, tarpaulins, and boat covers. It is produced in many grades and qualities. It may have a soft or firm hand. It is made in plain or basket weave. Canvas is smoother, more compact, and the heaviest of the three (Duck, canvas and sail cloth). It is tightly woven and very stiff plain weave fabrics made of even yarn for industrial use. They usually have an uneven weave pattern. Because of the tight weave, these fabrics are often used for outdoor purposes. It is made with single or ply yarns. Different types of canvas relate to which yarns (warp and weft) are plied and how many plies are used in the ply yarn. Duck is used for slipcovers, boat covers, shoe fabrics, house and store awnings, tarpaulins and covers for military and industrial uses.

Sail Cloth: It is a bottom weight half basket weave (2×1), unbalanced fabric of spun or textured filament yarns that can be piece dyed or printed. Sail cloth is the lightest (among the sail cloth, duck and canvas) in weight and made of single yarns. It is used in slacks, skirts, summer weight suits, and furnishings.

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Serge: It is a popular basic twill fabric made from any number of different fibers. When serge is made from wool, it is often woven from worsted yarns. Serge will take a crease well, but wool serge tends to become shiny with wear. It tailors well. Serge is a 2/2 twill with a subdued wale with combed or worsted yarns and a clear or hard finish (not napped or brushed). Serge with fine yarns, a high count, and a water-repellent finish is used for jackets, snowsuits, and raincoats. Heavier serge, with coarse yarns, is used for work pants. Serge often weighs 339.1 (10 oz/yd²) gm/m² or more. Fabric construction: Yarn size varies with fiber content 48×34 to 62×58.

Twill Flannel: It is a 2/2 twill. The weft yarns are larger low-twist woolen or worsted yarns, made especially for napping. Worsted flannels have less nap, take and hold a sharp crease better, show less wear, and sag less than woolen flannels. Even-sided flannel is used in apparel and upholstery. Fabric construction: Yarn size varies with fiber content 56×30 to 86×52.

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Shark skin: It is a 2/2 twill with a sleek appearance. It has a small-step pattern because both warp and weft yarns alternate one white yarn with one colored yarns. Shark skin is used primarily for slacks and suits.

Herring bone: These fabrics have the twill line reversed at regular intervals across the warp to produce a design that resembles the backbone of a fish, hence the name herring bone. Two different color yarns may be used to accentuate the pattern. Herring bone patterns can be very subtle or very pronounced. Herring bone is used in both apparel and furnishings. These are common in suiting fabrics.

Hound's tooth: It is a 2/2 twill fabric with a unique small eight-point pattern. Two yarns in contrasting colors in the warp and weft are used in groups of four to create the distinctive pattern. Hound's tooth fabrics also are used in apparel and furnishings.

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Denim: It is a cotton or cotton/polyester blend, durable heavy weight twill-weave, yarn-dyed fabric. Usually, the warp is colored and the weft is white. It is often a left-hand twill with a blue (indigo) warp and white weft for use in apparel in a variety of weights. Since it is a warp-faced twill, the colored warp yarns predominate on the face and the white weft yarns on the back. It is available in several weights, ranging from 203.46 gm/m² (6 oz/yd²) to 474.74 gm/m² (14 oz/yd²) or more in a 2/2 or 3/1 interlacing pattern. Its long-term popularity has made it a fashion fabric in casual wear. It may be napped, printed, made with spandex or other stretch yarns, or otherwise modified for fashion. Fabric construction: 7' to 16' x 8's to 23' / 60 x 36 to 72 x 44.

Drill: It is a strong, medium-to-heavy weight, warp-faced, twill weave fabric. It is usually a 2/2 or 3/1 left-handed twill and piece dyed (solid color). It is usually seen in work clothing and industrial fabrics.

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Jean: It is a warp faced twill of carded yarns. It is lighter weight than drill, and it has finer yarns but a higher warp-yarn count. It is usually with colored warp yarns and white weft yarns but can also be piece-dyed or printed medium-weight twill used for sportswear, draperies, slipcovers, and work shirts. Jean is not heavy enough for work pants. Fabric construction: 21' to 24' x 24' to 30' / 84 x 56 to 100 x 64.

Gabardine: It is a tightly woven, medium to heavy-weight, warp-faced steep or regular-angle, twill weave fabric with a very prominent, distinct wale that is closely set together and raised. It always has many more warp than weft yarns. It can be made of carded or combed single or ply yarns. The long-wearing fabric may be heather, striped, plaid, or solid color. The fabric can be wool, a wool blend, or synthetic fabrics that resemble wool. Gabardine can also be 100% texturized polyester or a cotton/polyester blend. Fabric construction: 15' to 39/2 x 15' to 26' / 110 x 76 to 130 x 80.

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Damask: Woven fabrics made from expensive mercerized cotton for bed clothing and table cloths. The figured design often is made by interchanging satin and sateen weaves.

Fil-à-fil: 2/2 twill weave fabrics in which light and dark colors alternate in both warp and weft, making a miniature staircase pattern. Used for suits and costumes.

OTHER COMMERCIAL FABRICS

Brocade: Heavily figured jacquard fabric, often with lusture yarn effects. Used in formal wear and furnishings.

Corduroy: A cut pile cord, usually in cotton. The cords may be of various widths. Used mainly in leisure and business wear.

Mull: Soft, fine, open, plain weave cotton fabric with a very low thread density in both warp and weft. Used for blouses and squares.

Pin Stripe: Worsted woven fabric with fine light-colored lines in the warp direction. Used for suits and costumes.

Ottoman: Warp-faced rib fabric with 3 to 10 ribs per cm for coats, jackets, and furnishings.

OTHER COMMERCIAL FABRICS

Panama: General term for a plain-based weave where two or more warp and weft yarns interlace as one, giving a chequered appearance. Made from cotton for shirts, tropical suits and leisure wear, or wool for suits and costumes.

Pocketing: Cotton plain woven fabric made smooth and dense by calendaring, for pocket linings.

Seersucker fabric: Cotton fabric with crinkled length-way stripes caused by differential shrinkage. True seersucker is generated by differential warp tensions but finishing treatments can produce a similar effect. Used for blouses, shirts, dresses.

Amazon: Very fine, satin weave for business suitings with fine worsted warp yarns and woolen weft. A light milling or raising finish may be given.

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Terry: A soft, voluminous, loop pile fabric. The loops are formed in a second warp sheet by a special weaving technique. Used for bath robes, towels, sports and leisure wear.

Whipcord: General term for densely woven warp-faced fabrics with a steep twill line. In worsted yarns with a clean finish for trousers, suits and coats.

Blazer Cloth: Traditionally an all-wool woven fabric for apparel, in either solid colors or stripes, that may be milled and/or raised. Imitation blazer cloths introduce cotton in the weft. The term may be used loosely for other fabrics for blazers.

Trouser: Trouser is a long pant or full pant.

FABRIC CONSTRUCTION

Name of the fabric	Construction or Specification	Weight (GSM)
Gabardine	$110 \times 51 / 20 \times 16 \mid 56''$	210
Gabardine	$72 \times 40 / 20/2 \times 10 \mid 56''$	270
Gabardine	$124 \times 70 / 20 + 20 \times 16 + 16 \mid 56''$	245
Gabardine	$108 \times 58 / 40/2 \times 40/2 \mid 56''$	200
Gabardine	$80 \times 48 / (10+10) \times 10 \mid 56''$	300
Gabardine	$116 \times 40 / 20 + 20 \times 7 \mid 56''$	285
Gabardine	$112 \times 54 / 16 + 16 \times 12 + 12 \mid 56''$	270
Twill fabric	$20 \times 7 / 128 \times 42 \mid 57'' \sim 58''$	298
Twill fabric	$7 \times 7 / 72 \times 40 \mid 57'' \sim 58''$	380
Twill fabric	$30 \times 30 / 130 \times 70 \mid 57'' \sim 58''$	160

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Slub Twill	$16 \text{ (slub)} \times 9 / 95 \times 48 \mid 57'' \sim 58''$	265
Herring bone Twill	$20 \times 20 / 108 \times 58 \mid 57'' \sim 58''$	195
Micro Twill	$150\text{D} \times 150\text{D} / 145 \times 80 \mid 57'' \sim 58''$	—
T.C. Twill	$45/2 \times 45/2 / 128 \times 70 \mid 57'' \sim 58''$	—
Cotton YD 2/1 Twill	$80/2 \times 100/2 / 142 \times 84 \mid 58'' \sim 59''$	—
Canvas	$10 \times 10 / 72 \times 42 \mid 57'' \sim 58''$	260
Canvas	$20 \times 16 / 100 \times 50 \mid 57'' \sim 58''$	190
Canvas Panama	$16+16 \times 12+12 / 108 \times 60 \mid 57'' \sim 58''$	—
Slub canvas	$16 \text{ (slub)} \times 9 / 80 \times 48 \mid 57'' \sim 58''$	260
Light canvas	$30 \times 30 / 108 \times 80 \mid 58'' \sim 59''$	—
Ribstop	$20 \times 20 / 108 \times 58 \mid 57'' \sim 58''$	195

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Ribstop	20 × 16 / 96 × 50 × 57 58"	193
Nylon Fabric	190T / 70D(24F) × 70D(24F) 60"	—
Polyester Fabric	95 × 80 / 70D × 75D 60"	—
Oxford	210T / 70D × 70D 58"	—
Polyester Concord	95 × 80 / 450D × 450D 58"	—
Mixed Oxford	45sp × 16 Cotton / 104 × 60 57" ~ 58"	—
Micro Shirting	75sp × 100D / 112 × 70 57" ~ 58"	—
Oxford (Yarn Dyed)	72 × 68 / 30 × 20 58" ~ 59"	—
Cotton Oxford	156 × 64 / 40 × 80/2 58" ~ 59"	—
Oxford	60 × 68 / 40/2 × 40/2 58" ~ 59"	—
Oxford (Yarn Dyed)	144 × 76 / 80/2 × 100/2 58" ~ 59"	—

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Sheeting	30 × 30 / 68 × 68 57" ~ 58"	105
Calico	16 × 16 / 60 × 60 57" ~ 58"	180 or 6.15 oz
Calico	14 × 14 / 64 × 58 57" ~ 58"	6.80 oz
Poplin	30 × 30 / 104 × 81 57" ~ 58"	—
Poplin	40 × 40 / 133 × 72 57" ~ 58"	—
Poplin	40/2 × 30 / 74 × 72 58" ~ 59"	—
Heavy Poplin	40/2 × 40/2 / 112 × 50 58" ~ 59"	—
T/C (65/35) Blended Poplin	45 × 45 / 133 × 72 58" ~ 59"	—
CVC (60/40) Blended Poplin	45 × 45 / 136 × 76 58" ~ 59"	—
Premium Poplin	60 × 50 / 110 × 76 58" ~ 59"	—
Premium Poplin	80/2 × 80 / 128 × 98 58" ~ 59"	—

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Premium Poplin	100/2 × 100/2 / 144 × 91 58" ~ 59"	—
Bedford Cord	07 × 07 / 72 × 42 57" ~ 58"	—
Bedford Cord	128 × 72 / 40 × 40 58" ~ 59"	—
Flannel (Printed)	42 × 44 / 24 × 13 42" ~ 43"	—
Twill Flannel (Y/D)	64 × 54 / 21 × 21 44" ~ 45"	—

THANK
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